

ATMA RAM SANATAN DHARMA
COLLEGE

UNIVERSITY OF DELHI



DATABASE MANAGEMENT
SYSTEMS
(PRACTICALS)

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INDEX

S.No.	TOPIC	PG. NO
1	Introduction Page	1
2	Index Page	2
3	Practicals List	5
4	Creation of Database	6
5	Putting Data	7
6	Practical - 1	8
7	Practical – 2	9
8	Practical – 3	10
9	Practical – 4	11
10	Practical – 5	12
11	Practical – 6	13
12	Practical – 7	14
13	Practical – 8	15
14	Practical - 9	16

16	Practical – 10	17
17	Practical – 11	18
18	Practical – 12	19
19	Practical – 13	20
20	Practical – 14	21
21	Practical – 15	22
22	Practical – 16	23
23	Practical – 17	24
24	Practical – 18	25
25	Practical – 19	26
26	Practical – 20, 21	27
27	Practical – 22	28
28	Practical – 23	29
29	Practical – 24	30
30	Practical – 25	31
31	Practical – 26	32
32	Practical – 27	33
33	Practical - 28	34

34	Practical – 29	35
35	Practical – 30	36
35	Practical - 31	37

Practicals List

- I. Create and use the following student-society database schema for a college to answer the given (sample) queries using the standalone SQL editor.**

STUDENT	<u>Roll No</u>	StudentName	Course	DOB
	Char(6)	Varchar(20)	Varchar(10)	Date

SOCIETY	<u>SocID</u>	SocName	MentorName	TotalSeats
	Char(6)	Varchar(20)	Varchar(15)	Unsigned int

ENROLLMENT	<u>Roll No</u>	<u>SID</u>	DateOfEnrollment
	Char(6)	Char(6)	Date

Here Rollno (ENROLLMENT) and SID (ENROLLMENT) are foreign keys.

- Retrieve names of students enrolled in any society.
- Retrieve all society names.
- Retrieve students' names starting with the letter 'A'.
- Retrieve students' details studying in courses 'computer science' or 'chemistry'.
- Retrieve students' names whose roll no either starts with 'X' or 'Z' and ends with '9'
- Find society details with more than N TotalSeats where N is to be input by the user
- Update society table for the mentor name of a specific society
- Find society names in which more than five students have enrolled
- Find the name of the youngest student enrolled in society 'NSS'
- Find the name of the most popular society (on the basis of enrolled students)
- Find the name of two least popular societies (on the basis of enrolled students)
- Find the students names who are not enrolled in any society.
- Find the students names enrolled in at least two societies
- Find society names in which maximum students are enrolled
- Find names of all students who have enrolled in any society and society names in which at least one student has enrolled
- Find names of students who are enrolled in any of the three societies 'Debating', 'Dancing' and 'Sashakt'.
- Find society names such that its mentor has a name with 'Gupta' in it.
- Find the society names in which the number of enrolled students is only 10% of its capacity.
- Display the vacant seats for each society.
- Increment Total Seats of each society by 10%
- Add the enrollment fees paid ('yes'/'No') field in the enrollment table.
- Update date of enrollment of society id 's1' to '2018-01-15', 's2' to the current date and 's3' to '2018-01-02'.
- Create a view to keep track of society names with the total number of students enrolled in it.
- Find student names enrolled in all the societies.
- Count the number of societies with more than 5 students enrolled in it
- Add column Mobile number in student table with default value '9999999999'
- Find the total number of students whose age is > 20 years.
- Find names of students who were born in 2001 and are enrolled in at least one society.
- Count all societies whose name starts with 'S' and ends with 't' and at least 5 students are enrolled in the society.
- Display the following information:
Society name Mentor name Total Capacity Total Enrolled Unfilled Seats

- II. Do the following database administration commands:**

Create user, create role, grant privileges to a role, revoke privileges from a role, create index

1. Creating student-society Database schema

```
1 • CREATE DATABASE studentSociety;
2 • USE studentSociety;
3 • CREATE TABLE STUDENT (
4     RollNo CHAR(6) PRIMARY KEY,
5     StudentName VARCHAR(20),
6     Course VARCHAR(10),
7     DOB DATE
8 );
9
10 • CREATE TABLE SOCIETY (
11     SocID CHAR(6) PRIMARY KEY,
12     SocName VARCHAR(20),
13     MentorName VARCHAR(15),
14     TotalSeats INT UNSIGNED
15 );
16
17 • CREATE TABLE ENROLLMENT (
18     RollNo CHAR(6),
19     SID CHAR(6),
20     DateOfEnrollment DATE,
21     FOREIGN KEY (RollNo) REFERENCES STUDENT(RollNo),
22     FOREIGN KEY (SID) REFERENCES SOCIETY(SocID)
23 );
```

Data Inside the Database

```
USE studentSociety;
```

```
INSERT INTO STUDENT (RollNo, StudentName, Course, DOB)  
VALUES
```

```
('S001', 'Amit Sharma', 'Computer', '2004-05-12'),  
( 'S002', 'Anita Verma', 'Economics', '2003-11-23'),  
( 'S003', 'Rahul Gupta', 'Physics', '2001-07-19'),  
( 'S004', 'Zara Khan', 'Computer', '2005-02-01'),  
( 'S005', 'Xavier Dsouza', 'Biology', '2002-09-30');
```

```
INSERT INTO SOCIETY (SocID, SocName, MentorName, TotalSeats)  
VALUES
```

```
('SC01', 'NSS', 'Dr. Mehta', 30),  
( 'SC02', 'Debating', 'Prof. Gupta', 25),  
( 'SC03', 'Dancing', 'Ms. Roy', 20),  
( 'SC04', 'Sashakt', 'Mr. Sharma', 15);
```

```
INSERT INTO ENROLLMENT (RollNo, SID, DateOfEnrollment)  
VALUES
```

```
('S001', 'SC01', '2025-01-15'),  
( 'S002', 'SC02', '2025-02-10'),  
( 'S003', 'SC01', '2025-03-05'),  
( 'S003', 'SC03', '2025-03-07'),  
( 'S004', 'SC02', '2025-04-01'),  
( 'S005', 'SC04', '2025-04-12');
```

Practical 1

Retrieve names of students enrolled in any society

Query:

```
USE studentSociety;  
SELECT DISTINCT s.StudentName  
FROM STUDENT s  
INNER JOIN ENROLLMENT e  
    ON s.RollNo = e.RollNo;
```

Output:

Result Grid			 Filter Rows:
	StudentName		
	Amit Sharma		
	Anita Verma		
	Rahul Gupta		
	Zara Khan		
	Xavier Dsouza		

Practical 2

Retrieve all society Names.

Query:

```
USE studentSociety;  
  
SELECT SocName  
FROM SOCIETY;
```

Output:

Result Grid			 Filter Rows:
	SocName		
▶	NSS		
	Debating		
	Dancing		
	Sashakt		

Practical 3

Retrieve students' names starting with the letter 'A'

Query:

```
USE studentSociety;  
SELECT StudentName  
FROM STUDENT  
WHERE StudentName LIKE 'A%';
```

Output:

Result Grid			 Filter Rows:
	StudentName		
	Amit Sharma		
	Anita Verma		

Practical 4

Retrieve students' details studying in courses 'computer science' or 'chemistry'.

Query:

```
USE studentSociety;  
SELECT *  
FROM STUDENT  
WHERE Course = 'Computer'  
       OR Course = 'Chemistry';
```

Output:

Result Grid   Filter Rows:				
	RollNo	StudentName	Course	DOB
▶	S001	Amit Sharma	Computer	2004-05-12
	S004	Zara Khan	Computer	2005-02-01
✖	NULL	NULL	NULL	NULL

Practical 5

Retrieve students' names whose Roll no. either starts with 'X' or 'Z' and ends with '9'

Query:

```
USE studentSociety;
SELECT StudentName
FROM STUDENT
WHERE RollNo LIKE 'X%'
      OR RollNo LIKE 'Z%';
```

Output:

Result Grid

Filter Rows:

StudentName

Practical 6

Find society details with more than N Total Seats where N is to be input by the user

Query:

```
USE studentSociety;  
SELECT *  
FROM SOCIETY  
WHERE TotalSeats > 20;
```

Output:

Result Grid   Filter Rows:				
	SocID	SocName	MentorName	TotalSeats
▶	SC01	NSS	Dr. Mehta	30
	SC02	Debating	Prof. Gupta	25
⊞	NULL	NULL	NULL	NULL

Practical 7

Update society table for the mentor name of a specific society

Query:

```
USE studentSociety;  
UPDATE SOCIETY  
SET MentorName = 'Piyush Das'  
WHERE SocID = 'SC02';  
  
SELECT *  
FROM SOCIETY  
WHERE SocID = 'SC02';
```

Output:

Result Grid   Filter Rows:				
	SocID	SocName	MentorName	TotalSeats
	SC02	Debating	Piyush Das	25
	NULL	NULL	NULL	NULL

Practical 8

Find society names in which more than five students have enrolled

Query:

```
USE studentSociety;  
SELECT s.SocName  
FROM SOCIETY s  
JOIN ENROLLMENT e  
      ON s.SocID = e.SID  
GROUP BY s.SocName  
HAVING COUNT(e.RollNo) > 5;
```

Output:

Result Grid			 Filter Rows
	SocName		

Practical 9

Find the name of the youngest student enrolled in society 'NSS'

Query:

```
SELECT s.StudentName
FROM STUDENT s
JOIN ENROLLMENT e
      ON s.RollNo = e.RollNo
JOIN SOCIETY soc
      ON e.SID = soc.SocID
WHERE soc.SocName = 'NSS'
ORDER BY s.DOB DESC
LIMIT 1;
```

Output:

Result Grid		Filter Rows:	
	StudentName		
▶	Amit Sharma		

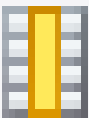
Practical 10

Find the name of the most popular society (on the basis of enrolled students)

Query:

```
SELECT s.SocName
FROM SOCIETY s
JOIN ENROLLMENT e
      ON s.SocID = e.SID
GROUP BY s.SocName
ORDER BY COUNT(e.RollNo) DESC
LIMIT 1;
```

Output:

Result Grid   Filter Rows:	
	SocName
	NSS

Practical 11

Find the name of two least popular societies (on the basis of enrolled students)

Query:

```
SELECT s.SocName
FROM SOCIETY s
JOIN ENROLLMENT e
      ON s.SocID = e.SID
GROUP BY s.SocName
ORDER BY COUNT(e.RollNo) ASC
LIMIT 2;
```

Output:

Result Grid			 Filter Rows:
	SocName		
	Sashakt		
	Dancing		

Practical 13

Find the students names enrolled in at least two societies

Query:

```
SELECT s.StudentName
FROM STUDENT s
JOIN ENROLLMENT e
      ON s.RollNo = e.RollNo
GROUP BY s.StudentName
HAVING COUNT(e.SID) >= 2;
```

Output:

Result Grid			 Filter Rows
	StudentName		
	Rahul Gupta		

Practical 14

Find society names in which maximum students are enrolled

Query:

```
SELECT s.SocName
FROM SOCIETY s
JOIN ENROLLMENT e
    ON s.SocID = e.SID
GROUP BY s.SocName
HAVING COUNT(e.RollNo) = (
    SELECT MAX(student_count)
    FROM (
        SELECT COUNT(RollNo) AS student_count
        FROM ENROLLMENT
        GROUP BY SID
    ) AS counts
);
```

Output:

Result Grid		Filter Rows:	
	SocName		
▶	NSS		
	Debating		

Practical 15

Find names of all students who have enrolled in any society and society names in which at least one student has enrolled

Query:

```
SELECT s.StudentName AS Name, 'Student' AS Type
FROM STUDENT s
JOIN ENROLLMENT e ON s.RollNo = e.RollNo
UNION
SELECT soc.SocName AS Name, 'Society' AS Type
FROM SOCIETY soc
JOIN ENROLLMENT e ON soc.SocID = e.SID;
```

Output:

Result Grid			Filter Rows:
	Name	Type	
▶	Amit Sharma	Student	
	Anita Verma	Student	
	Rahul Gupta	Student	
	Zara Khan	Student	
	Xavier Dsouza	Student	
	NSS	Society	
	Debating	Society	
	Dancing	Society	
	Sashakt	Society	

Practical 16

Find names of students who are enrolled in any of the three societies 'Debating', 'Dancing' and 'Sashakt'.

Query:

```
SELECT DISTINCT s.StudentName
FROM STUDENT s
JOIN ENROLLMENT e
    ON s.RollNo = e.RollNo
JOIN SOCIETY soc
    ON e.SID = soc.SocID
WHERE soc.SocName IN ('Debating', 'Dancing', 'Sashakt');
```

Output:

Result Grid		Filter	
	StudentName		
	Anita Verma		
▶	Rahul Gupta		
	Xavier Dsouza		
	Zara Khan		

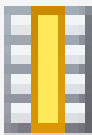
Practical 17

Find society names such that its mentor has a name with 'Gupta' in it

Query:

```
SELECT SocName  
  
FROM SOCIETY  
  
WHERE MentorName LIKE '%Gupta%';
```

Output:

Result Grid			 Filter Rows:
	SocName		

Practical 18

Find the society names in which the number of enrolled students is only 10% of its capacity

Query:

```
SELECT s.SocName
FROM SOCIETY s
JOIN ENROLLMENT e
    ON s.SocID = e.SID
GROUP BY s.SocID, s.SocName, s.TotalSeats
HAVING COUNT(e.RollNo) = (s.TotalSeats * 0.1);
```

Output:

[illegible]

Practical 19

Display the vacant seats for each society

Query:

```
SELECT s.SocName,  
       s.TotalSeats - COUNT(e.RollNo) AS VacantSeats  
FROM SOCIETY s  
LEFT JOIN ENROLLMENT e  
      ON s.SocID = e.SID  
GROUP BY s.SocID, s.SocName, s.TotalSeats;
```

Output:

Result Grid			Filter Rows:	
	SocName	VacantSeats		
▶	NSS	28		
	Debating	23		
	Dancing	19		
	Sashakt	14		

Practical 20

Increment Total Seats of each society by 10%

Query:

```
UPDATE SOCIETY  
SET TotalSeats = ROUND(TotalSeats * 1.1);
```

Practical 21

Add the enrollment fees paid ('yes'/'No') field in the enrollment table.

Query:

```
ALTER TABLE ENROLLMENT  
ADD FeesPaid ENUM('Yes', 'No');
```

Practical 22

Update date of enrollment of society id 's1' to '2018-01-15', 's2' to the current date and 's3' to '2018-01-02'

Query:

```
UPDATE ENROLLMENT
SET DateOfEnrollment = CASE SID
    WHEN 's1' THEN '2018-01-15'
    WHEN 's2' THEN CURRENT_DATE
    WHEN 's3' THEN '2018-01-02'
END
WHERE SID IN ('s1', 's2', 's3');
```

Practical 23

Create a view to keep track of society names with the total number of students enrolled in it

Query:

```
CREATE VIEW SocietyEnrollmentCount AS
SELECT s.SocName,
       COUNT(e.RollNo) AS TotalEnrolled
FROM SOCIETY s
LEFT JOIN ENROLLMENT e
      ON s.SocID = e.SID
GROUP BY s.SocID, s.SocName;
```

Practical 24

Find student names enrolled in all the societies

Query:

```
SELECT s.StudentName
FROM STUDENT s
JOIN ENROLLMENT e
      ON s.RollNo = e.RollNo
GROUP BY s.RollNo, s.StudentName
HAVING COUNT(DISTINCT e.SID) = (
      SELECT COUNT(*) FROM SOCIETY
);
```

Output:

Result Grid				Filter
	StudentName			

Practical 25

Count the number of societies with more than 5 students enrolled in it

Query:

```
SELECT COUNT(*) AS NumSocieties
FROM (
    SELECT s.SocID
    FROM SOCIETY s
    JOIN ENROLLMENT e
        ON s.SocID = e.SID
    GROUP BY s.SocID
    HAVING COUNT(e.RollNo) > 5
) AS sub;
```

Output:

Result Grid		Filter Rows:	
	NumSocieties		
▶	0		

Practical 26

Add column Mobile number in student table with default value '9999999999'

Query:

```
ALTER TABLE STUDENT  
ADD Mobile VARCHAR(15) DEFAULT '9999999999';
```


Practical 27

Find the total number of students whose age is > 20 years

Query:

```
SELECT COUNT(*) AS NumStudentsOver20  
FROM STUDENT  
WHERE TIMESTAMPDIFF(YEAR, DOB, CURDATE()) > 20;
```

Output:

Result Grid




Filter Rows:

	NumStudentsOver20
▶	5

Practical 28

Find names of students who were born in 2001 and are enrolled in at least one society

Query:

```
SELECT DISTINCT s.StudentName
FROM STUDENT s
JOIN ENROLLMENT e
    ON s.RollNo = e.RollNo
WHERE YEAR(s.DOB) = 2001;
```

Output:

Result Grid		Filter Rows	
	StudentName		
▶	Rahul Gupta		

Practical 29

Count all societies whose name starts with 'S' and ends with 't' and at least 5 students are enrolled in the society.

Query:

```
SELECT COUNT(*) AS NumSocieties
FROM SOCIETY s
JOIN ENROLLMENT e
    ON s.SocID = e.SID
WHERE s.SocName LIKE 'S%t'
GROUP BY s.SocID, s.SocName
HAVING COUNT(e.RollNo) >= 5;
```

Output:

Result Grid		Filter Rows:	
	NumSocieties		

Practical 30

Display the following information: Society name
Mentor name, Total Capacity, Total Enrolled,
Unfilled Seats

Query:

```
SELECT s.SocName,  
       s.MentorName,  
       s.TotalSeats AS TotalCapacity,  
       COUNT(e.RollNo) AS TotalEnrolled,  
       (s.TotalSeats - COUNT(e.RollNo)) AS UnfilledSeats  
FROM SOCIETY s  
LEFT JOIN ENROLLMENT e  
      ON s.SocID = e.SID  
GROUP BY s.SocID, s.SocName, s.MentorName, s.TotalSeats;
```

Output:

Result Grid  Filter Rows: Export:  Wrap Cell					
	SocName	MentorName	TotalCapacity	TotalEnrolled	UnfilledSeats
►	NSS	Dr. Mehta	30	2	28
	Debating	Piyush Das	25	2	23
	Dancing	Ms. Roy	20	1	19
	Sashakt	Mr. Sharma	15	1	14

II. Do the following database administration commands:

Create user, create role, grant privileges to a role, revoke privileges from a role, create index

Query:

```
CREATE USER 'newuser'@'localhost' IDENTIFIED BY 'password';  
CREATE ROLE manager_role;  
GRANT SELECT, INSERT, UPDATE ON student.* TO manager_role;  
REVOKE INSERT ON student.* FROM manager_role;  
CREATE INDEX idx_lastname ON STUDENT(StudentName);
```